Aryan kala BE24B004

Clue

Total residues in entire sequence (N) = 200

Total residues assigned to alpha helices (N_H) = 96

Methionine (M) Calculation

$N_M = 4$

$N_{M,H} = 3$

Property (P_M):

$$P_M = \frac{N_{M,H}/N_H}{N_M/N}$$

$$= \frac{3/96}{4/200} = \frac{0.03125}{0.02}$$

$$P_M = 1.5625 = 1.5625$$

Information Content (I_M):

$$I_M = \log_2(P_M)$$

$$I_M = \log_2(1.5625)$$

$$I_M \approx 0.6439$$

Proline (P) Calculation

$N_P = 9$

$N_{P,H} = 1$

$$\text{Property } (P_P) = \frac{N_{P,H}/N_H}{N_P/N} = \frac{1/96}{9/200}$$

$$= \frac{0.0104167}{0.045}$$

$P_P \approx 0.2315$

Aryankala BE24B004

Information Content I_p

$$I_p = \log_2(P_r)$$

$$I_p = \log_2(0.23148)$$

$$\boxed{I_p = -2.1110}$$

Que 4

(i) Manual Verification of Helix Segment: ELAAAMKRH

→ Initiation (6 residue window): Start at ELAAAM

assigning initiation scores E(1) + L(1) + A(1) + A(1) + A(1) + M(1)

= 6

As 6 ≥ threshold (4), a valid helix is initiated

→ Right Extension

Add K AAMK

Sum P_α = 1.45 + 1.45 + 1.20 + 1.07 = 5.17 ≥ 4

↳ Extend

Add R AMKR

Sum P_α = 1.45 + 1.20 + 1.07 + 0.79 = 4.51 ≥ 4

↳ Extend

Add H MKRH

Sum P_α = 1.20 + 1.07 + 0.79 + 1.24 = 4.30 ≥ 4

↳ Extend

Add G

→ KRHG

Sum P_α = 1.07 + 0.29 + 1.24 + 0.53 = 3.63 ≤ 4

Total prosperity

E(1.53) + L(1.34) + A(1.45) + A(1.45) + A(1.45)
+ M(1.20) + K(1.07) + K(0.79) +
H(1.24) ⇒ 11.520

This matches completely with our output of code.

(ii) Manual verification of strand segment KVFCRC

Initialization (Sliding window) : VFCRC

V(1) + F(1) + C(0) + R(0) + C(1) = 3 ≥ 3
↳ valid strand

threshold = 3

→ Left Extension

KVF

P_B = 0.74 + 1.65 + 1.28 = 3.67 ≥ 3
↳ Extend

→ Right Extension

RCE

P_B = 0.90 + 1.30 + 0.26 = 2.46 ≤ 3

extension terminates

Anupam Kedia
BE21B004

Total propensity

$$K(0.74) + N(1.05) + F(1.28) + G(0.81) + R(0.90) + C(1.50) = 6.680$$

Matches our data from code